

Exercise 3

```
[ec2-user@ip-172-31-20-184 lab8]$ mpicc -o pi_mpi pi_mpi.c -lm
mpirun --oversubscribe -np 4 ./pi_mpi
Rank 0: trials = 2500000, count = 1962662
Root received count = 1962547 from rank 1
Root received count = 1963912 from rank 2
Root received count = 1964200 from rank 3

[Pi] Processes = 4
[Pi] Trials = 10000000
[Pi] Points inside = 7853321
[Pi] Estimate of pi = 3.1413284000
[Pi] Error = 0.0002642536
[Pi] Time = 0.282547 sec
[ec2-user@ip-172-31-20-184 lab8]$ |
```

The 10,000,000 trials are divided between the processes (N / size each, the remainder goes to the first ranks). Each process seeds `rand()` with `SEED + rank` so that every process generates different points, counts how many of its points fall inside the quarter circle ($x^2 + y^2 \leq 1$) and sends only that count to root with `MPI_Send`. Root receives the counts in rank order with `MPI_Recv` (`source = r`) and calculates $\text{pi} = 4 \times \text{total_count} / N$.

With 4 processes 7,853,321 of the 10,000,000 points were inside, which gives $\text{pi} = 3.1413284$ (error = 0.00026). Time = 0.282547 sec. The estimate changes slightly with the number of processes because a different set of seeds is used.